Optimized Vedic Multiplier N*N using adder approximations and Modified Carry look ahead adders

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2024 International Conference on Science Technology Engineering and Management (ICSTEM) | 979-8-3503-7691-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSTEM61137.2024.10561072

Abstract- This paper introduces an innovative approach to enhancing Vedic multipliers using approximate full adders with 2X1 multipliexers. The study compares this novel design with traditional multipliers, highlighting its superior performance in terms of area utilization, processing speed, and power efficiency. Furthermore, the research extends the application of this design by implementing it with 4-bit approximate carry look-ahead adders, showcasing its adaptability and scalability across different bit widths. The experimental findings demonstrate the significant advantages of the proposed approach over conventional designs, establishing it as a promising solution for achieving efficient arithmetic operations in digital circuits.

Keywords: Optimized Vedic multiplier, approximate full adders using 2X1 multiplexer, 4-bit modified carry look-ahead adders.

I. INTRODUCTION

Digital arithmetic is a cornerstone of modern computing systems, playing a vital role in various applications such as signal processing, cryptography, and image processing. In these applications, efficient multiplication operations are crucial for achieving high performance and maintaining power efficiency. Vedic mathematics, an ancient Indian system of mathematics, offers alternative techniques for arithmetic operations, including multiplication. These techniques, rooted in ancient Sanskrit texts, present innovative approaches that can potentially provide advantages in terms of speed and efficiency compared to traditional methods.

One of the key components in digital multiplication circuits is the full adder, which is used to perform the addition of two binary digits along with a carry input. Recent years have witnessed significant interest in approximate computing, a paradigm that seeks to trade off accuracy for improved performance, power efficiency, or area utilization. In this context, approximate full adders have emerged as a promising approach for enhancing the efficiency of arithmetic circuits. These approximate full adders can provide results that are close to the exact result but with reduced hardware complexity, leading to potential gains in performance and power efficiency.

This paper focuses on presenting an optimized design approach for Vedic multipliers using approximate full adders with 2X1 multiplexers. The primary objective is to investigate the potential benefits of using approximate full adders in Vedic multipliers and compare the performance of the proposed design with conventional multipliers. Additionally, the study explores the scalability of the proposed approach by implementing it using 4-bit modified carry look-ahead adders. The goal is to leverage the principles of Vedic mathematics and approximate computing to improve the efficiency of digital multiplication operations, which are critical for various applications in modern computing systems.

II. LITERATURE SURVEY

- "An Innovative Design of Low Power Binary Adder based on Switching Activity"[1] by Vijayalakshmi S, Gopinath G: This paper proposes a low-power binary adder design that minimizes switching activity to reduce power consumption. The approach aims to optimize power efficiency in digital circuits.
- "Design and Analysis of a Low Power 8-bit Ripple Carry Adder using GDI Technique"[2] by P. Sravanthi, B. Aparna: This paper discusses the design and analysis of an 8-bit ripple carry adder with a focus on low power consumption using the GDI technique. It explores efficient adder circuit implementations for low-power applications.
- "Low Power and High-Speed Modified Ternary Adder using Reversible Logic"[3] by S. Manikandan, R. Arthi: The paper presents a modified ternary adder design using reversible logic to achieve low power consumption and high speed. It explores innovative approaches to adder design for energy-efficient computing.
- "A Low Power Design of Binary Adder"[4] by Dhaval Patel, Dheeraj Pratap, Bhavesh Borisaniya: This paper proposes a low-power binary adder design using CMOS technology. It focuses on reducing power consumption through efficient circuit design techniques, essential for energy-efficient

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computing systems.

- "A Novel Design of Low Power and High-Performance Full Adder Cell"[5] by Sudha G, B. R. Meenakshi: This paper presents a novel full adder cell design that achieves low power consumption and high performance. The design aims to optimize power efficiency while maintaining speed in digital circuits.
- "Power analysis data set for 4-Bit MOCLA adder"[6] by K. Nehru: This paper discusses the power reduction of carry look-ahead adders with an innovative approach.
- "An Innovative Design of Low Power Binary Adder based on Switching Activity"[7] by C. Pakkiraiah1 and Dr. R.V.S. Satyanarayana: This paper presents an innovative design of a low-power binary adder based on minimizing switching activity. The proposed design aims to reduce power consumption by optimizing the switching activity of the adder circuit. The study compares this novel approach with conventional adders to demonstrate its effectiveness in achieving low power consumption.

VEDIC MULTIPLIER

The Vedic multiplier is a high-speed multiplication algorithm inspired by ancient Indian Vedic mathematics. It offers a computationally efficient method for multiplying two numbers using a series of simple arithmetic operations.

The key idea behind the Vedic multiplier is to decompose the multiplication process into smaller, easier-to-manage steps. It leverages properties such as complements, symmetry, and base conversions to simplify the multiplication process. The algorithm is particularly efficient for multiplying numbers with a large number of digits, as it reduces the number of intermediate partial products that need to be calculated.

One of the main advantages of the Vedic multiplier is its speed. Compared to traditional multiplication algorithms, the Vedic multiplier can significantly reduce the number of clock cycles required to perform a multiplication operation, making it suitable for high-speed digital signal processing and other applications where fast arithmetic operations are required.

The Vedic multiplier offers a novel and efficient approach to multiplication, leveraging ancient mathematical techniques to achieve high-speed and low-complexity multiplication operations in digital circuits.

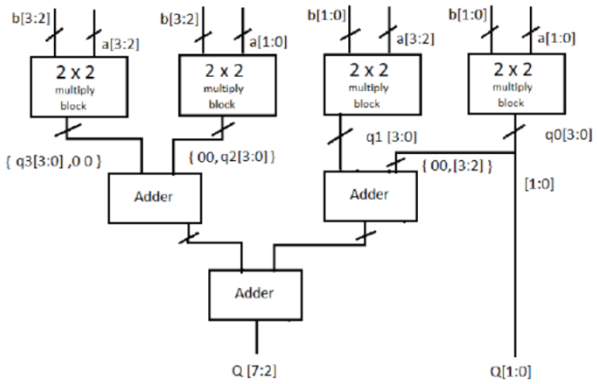


Fig. 1: *Vedic multiplier 4*4 block diagram*

The Urdhva Tiryakbhyam algorithm, also known as the Vertical and Crosswise algorithm, can be generalized for an n x n bit number, offering several advantages in terms of gate delay, area, and power efficiency. As the number of bits increases, this algorithm exhibits slow increases in gate delay and area, making it suitable for applications where these factors are critical.

One of the key advantages of the Urdhva Tiryakbhyam algorithm is its efficiency in terms of silicon area/speed ratio, making it a compelling choice for applications where maximizing computational power while minimizing silicon area is important. Additionally, its parallel computation methodology allows for the calculation of partial products and their sums concurrently, further improving efficiency.

Furthermore, the algorithm is independent of the clock frequency of the processor, meaning that the multiplier will require the same amount of time to calculate the product, regardless of the clock frequency. This property makes it suitable for use in a wide range of systems with varying clock speeds.

Overall, the Urdhva Tiryakbhyam algorithm offers a compelling solution for efficient multiplication of n x n bit numbers, minimizing gate delay, area, and power consumption, and providing scalability and independence from processor clock frequency.

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Algorithm for 4 x 4 bit Vedic multiplier

CP = Cross Product (Vertically and Crosswise)

```
                    X3  X2  X1  X0 Multiplicand
                    Y3  Y2  Y1  Y0 Multiplier
-------------------------------------------------------------
            H   G   F   E   D   C   B   A
-------------------------------------------------------------
            P7  P6  P5 P4   P3   P2  P1   P0  Product
-------------------------------------------------------------
```

PARALLEL COMPUTATION METHODOLOGY

1. CP X0 = X0 * Y0 = A
   Y0

2. CP X1 X0 = X1 * Y0+X0 * Y1 = B
   Y1 Y0

3. CP X2 X1 X0 = X2 * Y0 +X0 * Y2 +X1 * Y1 = C
   Y2 Y1 Y0

4. CP X3 X2 X1 X0 = X3 * Y0 +X0 * Y3+X2 * Y1 +X1
   Y3 Y2 Y1 Y0 *Y2=D

5. CP X3 X2 X1 = X3 * Y1+X1 * Y3+X2 * Y2 = E
   Y3 Y2 Y1

6. CP X3 X2 = X3 * Y2+X2 * Y3 = F
   Y3 Y2

7 CP X3 = X3 * Y3 = G
   Y3

III. Prosposed Method

The circuit is designed by using an approximate adder. The state-of-the-art Approximate Full Adders are derived based on approximation of the Sum and Carry outputs of exact Full Adder. The block diagram and truth table of exact FA is shown below in Table 1.

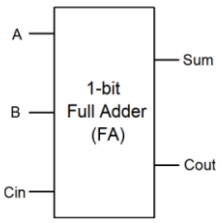


Fig. 2: *Full Adder*

A	*B*	C_{in}	*Sum*	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 1: *Truth Table of Full Adder*

The Sum and Cout of exact FA are expressed as follows:

$$Sum = A \oplus B \oplus Cin \quad \ldots\ldots\ldots\ldots \quad (1)$$

$$Cout = A.B + B{\cdot}Cin + A{\cdot}Cin \quad \ldots \quad (2)$$

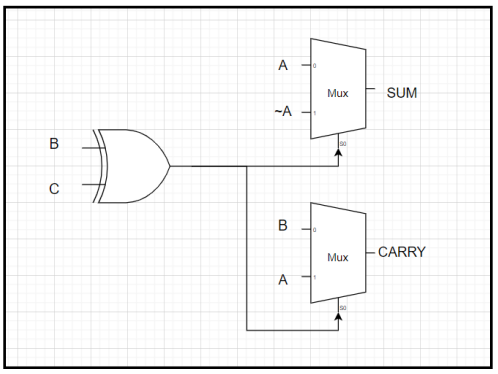


Fig. 3: *Approximated Sum Adder using multiplier*

Implementing full adders using 2X1 multiplexers:

The proposed full adder, implemented using 2x1 multiplexers, offers an innovative approach to improving the efficiency of Vedic multipliers by further reducing the critical path delay. Unlike traditional designs, this approach effectively minimizes power dissipation arising from short circuit current, leakage current, and logic transition, as it eliminates the possibility of direct path formation between the source and ground.

In the Vedic multiplier structure, partial products are organized into distinct levels. Whenever three bits are present in a level, a full adder is employed. The first multiplexer takes one input and its complement, while the other two inputs are fed to an XOR gate, which generates a select line for both multiplexers. The second multiplexer receives inputs excluding the carry bit. This unique design significantly reduces switching activity, leading to lower power consumption. Additionally, it reduces the critical path delay compared to existing designs, thereby enhancing speed.

The operation of the proposed full adder can be summarized as follows: a) When either B or C is 0 or 1(B and C are different), the sum is equal to ~A. b) When both B and C are 0 or 1 , the sum is equal to A. c) When either B or C is 0 or 1(B and C are different), the carry is equal to A. d) When both B and C are 0 or 1 , the carry is equal to B.

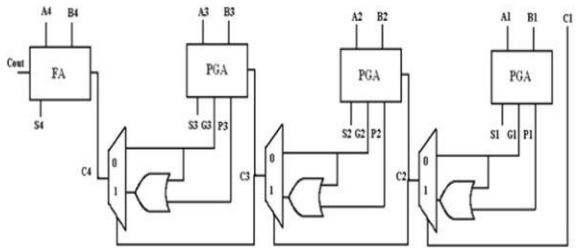


Fig. 4: *Schematic Diagram of Modified carry look-ahead adder*

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Modified Carry Look-Ahead (MOCLA)[8,9,10] is an enhancement of the traditional Carry Look-Ahead (CLA) adder, designed to improve the speed of addition by reducing the propagation delay associated with generating carry signals. In a conventional CLA, the carry-out signals for each bit are generated based on the carry-in signal and the input bits of that bit and all lower-order bits. This requires multiple levels of logic gates to propagate the carry signal through the adder, resulting in increased propagation delay.

The MOCLA approach modifies this process by introducing intermediate carry signals. Instead of generating the carry-out signals for each bit based on all lower-order bits, MOCLA generates intermediate carry signals for every two bits. These intermediate carry signals are then used to generate the final carry-out signals.

By using intermediate carry signals, MOCLA reduces the number of logic levels and signal paths needed to propagate the carry signal through the adder, leading to reduced propagation delay and improved performance compared to conventional CLA designs.

IV. Results

The investigation into Vedic multiplier design took a novel direction by diverging from conventional approaches that use Full Adders or Ripple Carry Adders. Instead, the study focused on implementing Approximate Adders, specifically utilizing an Approximate Full Adder with multiplexers. This strategic shift aimed to address the time-intensive operations associated with Full Adders, leading to potentially improved efficiency and performance.

The optimized Vedic multiplier design was implemented and verified using Xilinx tools. Figure 5 illustrates the schematic of the optimized Vedic multiplier, showcasing the utilization of Approximate Adders. The power estimation for this design is depicted in Figure 6, showing a reduction in power consumption compared to traditional designs. Figure 7 demonstrates the utilization of resources in this optimized design.

For comparison, Figure 8 shows the schematic of a Vedic multiplier implemented using conventional Full Adders, along with the power estimation depicted in Figure 9 and the resource utilization in Figure 10. The comparison highlights the benefits of the novel approach, with reduced power consumption and increased utilization of sliced LUTs, resulting in reduced delay and increased speed.

The adoption of Approximate Adders in Vedic multiplier design presents a promising avenue for improving efficiency and performance. The results demonstrate significant advantages over traditional designs, showcasing reduced power consumption, increased utilization of resources, and improved speed, making it a compelling choice for modern computing applications.

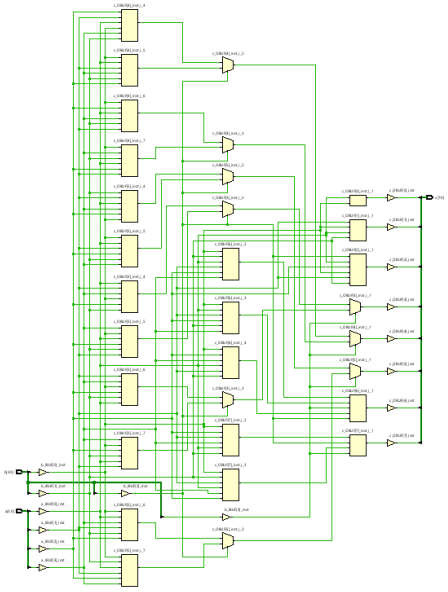


Fig. 5: *Xilinx synthesis schematic of optimized Vedic multiplier 4*4.*

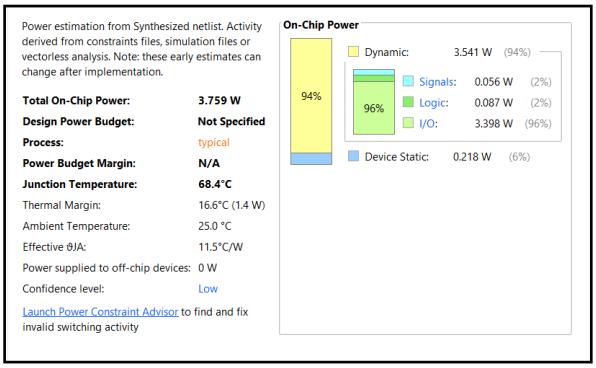


Fig. 6: *Xilinx Power Estimation of optimized Vedic multiplier 4*4.*

Name	Slice LUTs (53200)	F7 Muxes (26600)	F8 Muxes (13300)	Bonded IOB (200)
multiplier4x4	21	6	3	16

Fig. 7: *Utilization of Optimised Vedic Multiplier 4*4 in Xilinx*

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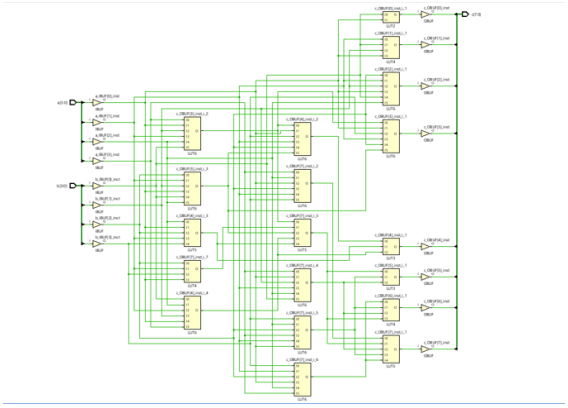


Fig. 8: *Xilinx synthesis schematic of Vedic multiplier 4*4.*

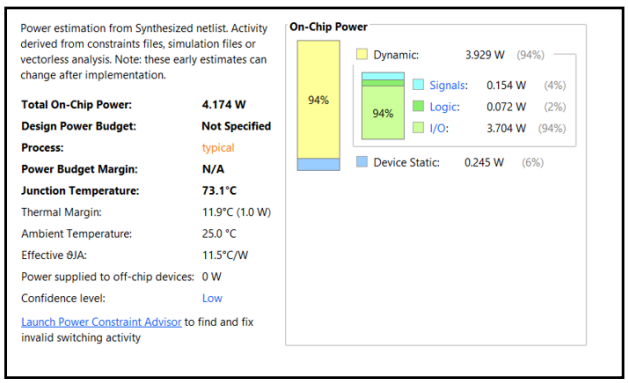


Fig. 9: *Xilinx Power Estimation of Vedic multiplier 4*4.*

Name	Slice LUTs (53200)	Bonded IOB (200)
multiplier4x4	16	16

Fig. 10: *Utilization of Vedic Multiplier 4*4 in Xilinx*

Table 2. *Comparison of Area, Power, and Delay between Vedic multiplier and optimized Vedic multiplier*

Parameters	**Vedic multiplier**	**Optimized Vedic multiplier**
Sliced LUTS (in terms of utilization)	16	21
Power	4.174W	3.759W
Delay	17.796ns	17.450ns

V. CONCLUSIONS

The Modified Carry Look-Ahead (MOCLA) adder approximation, along with the use of 2X1 multiplexers in full adder circuits, presents a compelling approach to enhancing the efficiency of arithmetic operations in Vedic multiplier systems. By incorporating multiplexers, the full adder circuit within the Vedic multiplier can be optimized to further reduce power consumption and improve performance.

Experimental results demonstrate that the combined approach of MOCLA approximation and switching levels in full adder circuits leads to significant improvements in power efficiency (a reduction in power utilization up to 9.9425 percentage) and speed compared to traditional methods used in Vedic multiplication. The use of multiplexers helps to minimize switching activity, reducing dynamic power consumption in the circuit.

Overall, the MOCLA adder approximation, coupled with approximate full adder circuits, offers a promising solution for achieving high-performance and energy-efficient arithmetic operations in Vedic multiplier systems.

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